

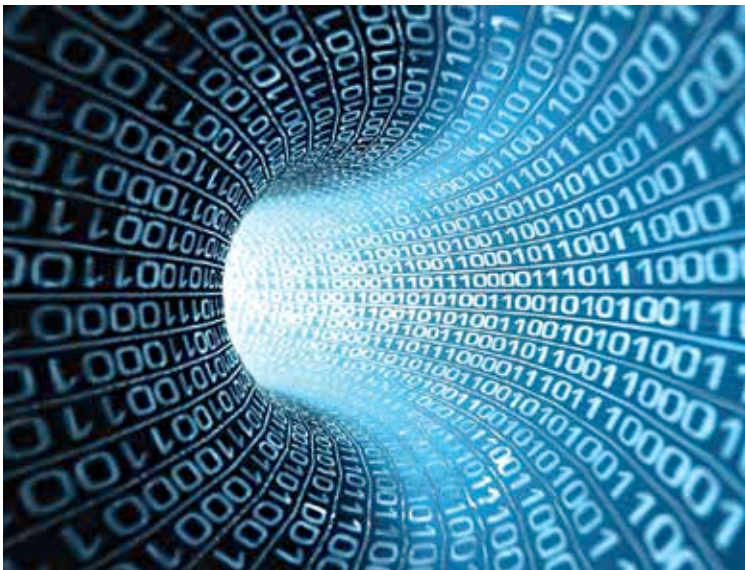
While it is just to assume that scientists would be overjoyed with this explosion of data, application of big data in science is not without its challenges. In fact, before taking a closer look at the applications themselves, it'll be beneficial for the reader to have at least a cursory knowledge of the challenges so that s/he can better appreciate the practicalities of analysing big data for practical scientific applications.

The challenges

1. Understanding huge chunks of data

The past decade has seen an overwhelming advancement in terms of data storage facilities-something that makes storing of huge volumes of data not just feasible but also easy. Due to massive storage facilities computer scientists are amassing data like never before. But collecting and storing information is quite distinct from gleaning understandings from it. The question of how to interpret this large database becomes significant, and a significant brain teaser at that.

The traditional statistical tests as well as computing models for making scientific inferences were created to analyze very small data samples garnered from relatively large populations. These models by their nature



Scientific experiments can be of a massive scale, generating a lot of data